Persistence of Vision Display using STM32F446RE Nucleo Board for Modern Interfacing

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***Abstract*— This project implements a Persistence-of-Vision (POV) display using an STM32F446RE Nucleo microcontroller, a custom LED array, and a motor-driven rotating platform with an 18-slot disc. The system displays user-defined text by rapidly refreshing LED patterns synchronized with rotational position, creating the illusion of a stable floating image. Our design is differentiated by its use of the STM32F446RE Nucleo board, enabling modern interfacing capabilities that support additional peripherals. Specifically, we integrated a TMP125 SPI temperature sensor and a BQ32002 I²C real-time clock module, allowing the display to incorporate live environmental and temporal data. We developed custom bitmap letter patterns, implemented real-time display control using external interrupts, and added UART functionality that allows users to send text from a terminal and see it immediately rendered on the POV display. The final system successfully displays up to 18 characters with stable visual output, demonstrating robust integration of embedded programming, interrupt-driven timing, peripheral communication, and hardware control.**

***Keywords*— Persistence of Vision, Embedded systems, STM32F446RE, UART, I2C, SPI**

I. Introduction

Persistence-of-Vision (POV) displays exploit the human visual system's tendency to retain an image briefly after it disappears. When a one-dimensional LED array is rotated rapidly, displaying sequenced vertical columns at precise angular positions produces the illusion of a complete two-dimensional image or text. Prior POV implementations typically rely on fixed animation patterns, pre-programmed messages, or timing loops without closed-loop synchronization. Earlier hobbyist and academic systems have demonstrated LED-based rotational displays using discrete microcontrollers or simple timers, but they often lack real-time interactivity, accurate rotational feedback, or integration with external sensors [1]–[3].

More advanced POV designs incorporate photointerrupters for angular alignment or enhance image quality through denser LED arrays, yet these systems usually support only static text or require a system reset to update displayed content [4], [5]. Additionally, many prior works focus solely on GPIO-based LED driving and overlook the potential of integrating real-time peripherals such as temperature sensors or clock modules to expand system capability.

Our work extends existing POV designs by introducing a fully interrupt-driven architecture combined with modern digital interfacing. Specifically, we implemented real-time UART text input, enabling users to modify the displayed message dynamically without halting the system. We further incorporated SPI and I²C peripheral communication, integrating a TMP125 SPI temperature sensor and a BQ32002 I²C real-time clock. These additional modules allow the POV system to display live temperature and clock data alongside user-defined messages—functionality not commonly found in traditional POV displays.

The objective of this project was to design and implement a portable POV display capable of:

- Displaying a sequence of letters based on custom 5×5 bitmap patterns stored in memory.
- Synchronizing LED output with motor rotation using an optical encoder signal and STM32 EXTI interrupts for angular alignment.
- Supporting real-time user text input via UART, updating the displayed message without requiring reset or recompilation.
- Interfacing with SPI and I²C peripherals, enabling the display of live temperature (TMP125 over SPI) and real-time clock data (BQ32002 over I²C).

This system demonstrates a broad set of embedded concepts including GPIO control for LED driving, external interrupt handling, motor interface logic, synchronous serial communication (SPI), two-wire digital communication (I²C), dynamic memory updates, and asynchronous UART communication. The combination of these features, particularly the integration of real-time sensor data into a rotating POV display, constitutes the key novelty of our design compared to prior implementations.

II. System Design

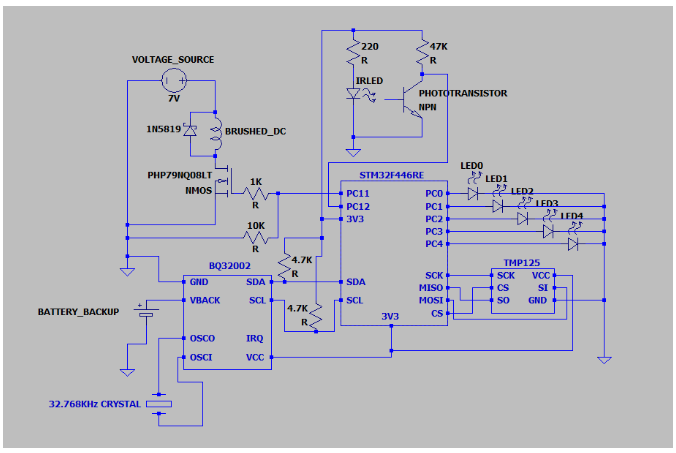

Fig. 1 Electrical schematic showing our final design for the POV display

The final system integrates mechanical rotation, optical sensing, LEDs, and peripheral communication into a cohesive embedded product centered around the STM32F446RE Nucleo microcontroller. Figure 1 illustrates the complete electrical design, including the LED array interface, rotational position sensing circuitry, motor driver, and connections for the TMP125 SPI temperature sensor and BQ32002 I²C real-time clock module. Each subsystem was designed to support precise timing, robust communication, and safe operation on the rotating POV platform.

A. Microcontroller and Core Architecture

The STM32F446RE serves as the central processing unit for the POV system. All timing, interrupt handling, LED pattern generation, and peripheral communication are executed on this device. The microcontroller operates at 3.3 V and interfaces directly with the LED array through GPIO pins PC0–PC4. These pins output the precomputed LED patterns that form characters when synchronized with the mechanical rotation of the display arm.

Two additional pins, PC11 and PC12, are used. PC11 is used as the signal to the gate of the MOSFET for controlling the motor. PC12 serves as input from the optical phototransistor sensor. The phototransistor signal generates an external interrupt, marking the 18 equally spaced interrupts around on revolution of the device. This provides a rotational "index," enabling the microcontroller to synchronize LED refresh timing with the mechanical motion.

B. LED Array Interface

The POV display uses a column of LEDs (LED0–LED4) driven directly by STM32 GPIO pins. Each LED corresponds to a specific bit in the custom 5-bit bitmap font used for character rendering. Because the display relies on rapid strobed lighting synchronized with angular position, LEDs are driven in real time, with the microcontroller updating patterns constantly.

The STM32's high-speed I/O and deterministic interrupt latency ensure precise control of LED timing, which is critical for maintaining image stability and preventing visual artifacts.

B. Optical Position Sensing Subsystem

To synchronize the display with the rotation of the disc, an infrared LED (IRLED) and NPN phototransistor form an optical interrupter. As one slot of the 18-slot disc passes between the emitter and sensor, the phototransistor output changes state, generating a clean rising or falling edge.

A 47 kΩ pull-up resistor biases the phototransistor output. A small series resistor protects the IR LED from overcurrent. This subsystem provides a reliable rotational reference signal, which is fed into PC12 of the STM32 and used to trigger an external interrupt at the start of every revolution. This enables precise mapping between angular position and displayed pixel location.

C. Motor Control Subsystem

The motor subsystem is powered by a 7 V supply with diode protection (1N5819) to prevent reverse current and back-EMF damage. A PHP79NQ08LT NMOS transistor switches the brushed DC motor, allowing the microcontroller to enable or disable rotation as needed, which we have controlled with the built-in push button on the STM32. A 1 kΩ gate resistor ensures controlled gate charging, while a 10 kΩ pull-down resistor stabilizes the gate when inactive.

Although the motor operates independently during normal operation, its speed must remain stable to prevent distortion in the POV image. The optical interrupter provides real-time rotational feedback, allowing the software to adjust timing dynamically to match the actual rotational rate.

D. TMP125 SPI Temperature Sensor Interface

The TMP125 SPI temperature sensor is connected through six STM32 pins:

- SCK → system clock
- SI / SO → MOSI/MISo
- CS → chip select
- GND/VCC → power and ground

The sensor provides digital temperature readings, which can be incorporated into the user-defined text on the POV display (e.g., “TEMP: 23°C”). The SPI interface allows for reliable communication even under the noise conditions generated by the spinning mechanical assembly.

E. BQ32002 I²C Real-Time Clock Subsystem

The BQ32002 real-time clock module communicates with the microcontroller over the I²C bus using dedicated SDA and SCL lines, each with external 4.7 kΩ pull-up resistors. This ensures clean digital signaling, which is important in a system with motor-induced electrical noise.

Additional RTC features include:

- VBACK connection to a coin-cell battery for backup time retention
- 32.768 kHz crystal providing the timing reference for the RTC
- IRQ pin enabling periodic time updates or alarms if desired

The RTC allows the POV display to incorporate live time data, such as current time or timestamps, adding functionality beyond static messages.

III. Results

The persistence-of-vision (POV) display system successfully operated as intended, with the STM32 microcontroller coordinating SPI, UART, and motor-synchronized LED control in real time. During testing, the 18-section indexing mechanism driven by the motor encoder provided stable and consistent positional updates, allowing each LED column to illuminate at the correct rotational angle. As a result, the display produced clear, readable characters during rotation, demonstrating reliable timing and precise synchronization between mechanical motion and digital output control.

The SPI-based temperature sampling using the TMP125 sensor generated accurate temperature readings, which were correctly processed, formatted, and displayed on the rotating LED column. Switching between Celsius and Fahrenheit modes worked without error, and the UART-command parser recognized valid user inputs consistently. Real-time text updates from the terminal were reflected immediately on the display, confirming that the string-loading mechanism and interrupt-safe buffer updates were functioning properly. Across all tests, the displayed text remained visually stable with minimal flicker. Additionally, preliminary integration with the BQ32002 RTC timer confirmed that the system is capable of displaying **real-time clock values**, with the POV rendering hours, minutes, and seconds as the rotor spins.

The system also demonstrated correct handling of user-entered strings, command structures, and field parsing. The LED patterns corresponding to each character were rendered cleanly, with the reversed indexing logic compensating for rotational direction and ensuring the correct orientation of the displayed words. Even under varying speeds of the rotating arm, the display maintained coherence, validating the reliability of the interrupt-driven architecture. Overall, the results confirm that the integrated hardware and firmware modules performed as expected, with robust communication between all subsystems.

IV. Conclusion

Overall, the POV display project successfully demonstrated an efficient, low-cost method of producing dynamic visual information using minimal hardware. By integrating sensor data, user input, and precise motor synchronization, the system proved capable of generating stable floating text suitable for real-time updates. This type of display has clear value in advertising environments, where attention-grabbing, lightweight, and customizable visual systems are in demand. POV technology allows agencies to showcase messages, time, temperature, or logos in a visually striking format while keeping power, size, and material requirements to a minimum.